EXPERIMENT NO: 02

30/7/2026

STUDY OF DIFFERENT TYPES OF NETWORK CABLES

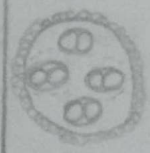
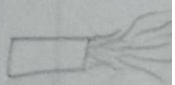
Aim: To study different types of Network Cables.

a) Understand different types of network cable

Different types of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable

Cable Type	Category	Maximum Data Transmission

Cable Type	Category	Maximum Data Transmissio	Advantages/ Disadvan -tages	Use	Image
UTP	Category 3	10 bps	Advantages • Cheaper in cost • Easy to install as they have a smaller overall diameter Disadvantages • More prone to EMI & noise	10Base-T Ethernet	
	Category 5	Upto 100 Mbps		Fast Ethernet, Gigabit Ethernet	
	Category 5e	1Gbps		Fast Ethernet, Gigabit Ethernet	
STP	Category 6, 6a	10Gbps	Advantages • Shielded • Faster than UTP • Less susceptible to noise & interference Disadvantages • Expensive • Greater installat° effort	Gigabit ethernet, 10G ethernet (55m) widely used in data centres	 
SSTP	Category 7	10Gbps		Gigabit ethernet, 10G ethernet (100m)	
Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages High bandwidth Immune to interference	Speed of signal is 500m TV network	

Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	• Low loss bandwidth • Versatile Disadvantages • Limited distance • Cost • Size is bulky	High Speed internet connect-ions
fibre optics cable	Single mode Multi mode	100Gbps	Advantages • High speed • High bandwidth • High Security • Long Distance Disadvantages ~~Limited distance~~ ~~Cost~~ • Expensive • Requires skilled installers	Maximu-m distance of fibre optics cable is around 100m

Result: Thus, different types of network cables were studied.